

WeHelp
Assignment - Week 2

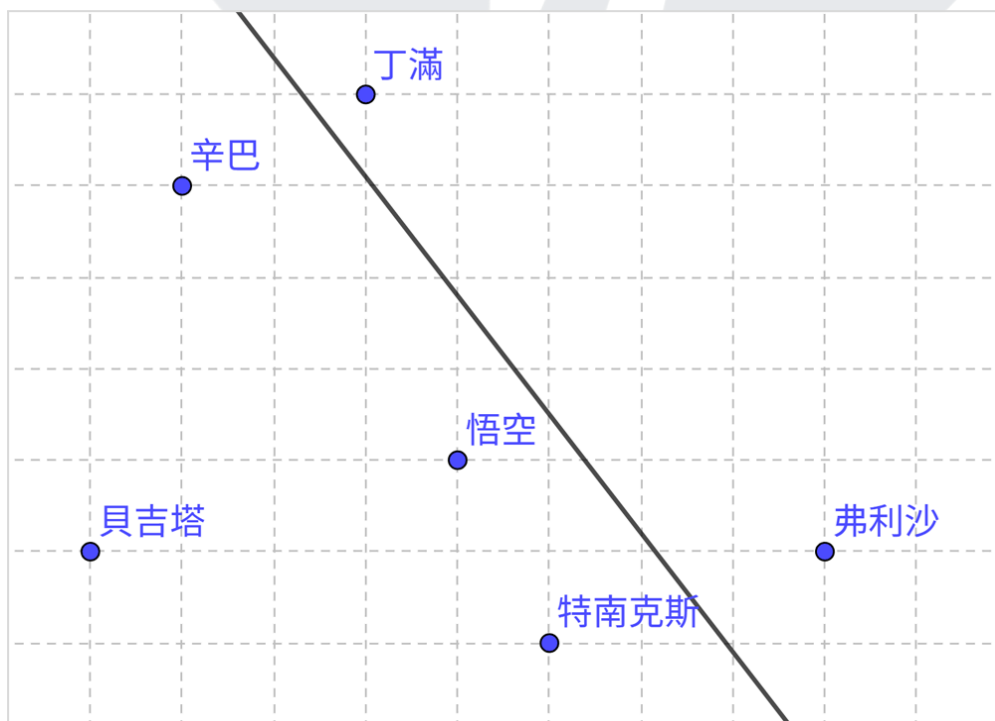
Complete tasks by Python and JavaScript without third party libraries.

Note: Python code should run on Python 3 or above version.

Task 1:

Based on the map shown below. Given any character name, find the characters are farthest and closest from the given one.

- Assume that coordinate of 悟空 is (0, 0).
- Assume that each cell is 1 distance unit.
- Characters move only horizontally and vertically.
- If a character moves to the other side, add additional 2 distance units.
- If there are multiple characters at the same distance, print them all.



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Python

```
def func1(name):  
    # your code here  
func1("辛巴") # print 最遠弗利沙；最近丁滿、貝吉塔  
func1("悟空") # print 最遠丁滿、弗利沙；最近特南克斯  
func1("弗利沙") # print 最遠辛巴，最近特南克斯  
func1("特南克斯") # print 最遠丁滿，最近悟空
```

JavaScript

```
function func1(name){  
    // your code here  
}  
func1("辛巴"); // print 最遠弗利沙；最近丁滿、貝吉塔  
func1("悟空"); // print 最遠丁滿、弗利沙；最近特南克斯  
func1("弗利沙"); // print 最遠辛巴，最近特南克斯  
func1("特南克斯"); // print 最遠丁滿，最近悟空
```

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Task 2:

Assume we have 1-on-1 services for booking and each service can serve 1 client at a time. Help people book the best matching service in a day, based on start hour, end hour, and selection criteria.

1. Booking requests are one by one, order matters.
2. Criteria is in one of the formats:"Field=Value", "Field>=Value", and "Field<=Value".
3. For field of service name, criteria only use the equal operator. For other fields, criteria only use those unequal operators.
4. Choose the available service which best matches the criteria. And print it.
5. If there are no available services, print "Sorry".

Note: Never change existing code.

Python

```
# your code here, maybe
def func2(ss, start, end, criteria):
    # your code here
services=[
    {"name":"S1", "r":4.5, "c":1000},
    {"name":"S2", "r":3, "c":1200},
    {"name":"S3", "r":3.8, "c":800}
]
func2(services, 15, 17, "c>=800") # S3
func2(services, 11, 13, "r<=4") # S3
func2(services, 10, 12, "name=S3") # Sorry
func2(services, 15, 18, "r>=4.5") # S1
func2(services, 16, 18, "r>=4") # Sorry
func2(services, 13, 17, "name=S1") # Sorry
func2(services, 8, 9, "c<=1500") # S2
```

JavaScript

```
// your code here, maybe
function func2(ss, start, end, criteria){
    // your code here
}

const services=[
    {"name":"S1", "r":4.5, "c":1000},
    {"name":"S2", "r":3, "c":1200},
    {"name":"S3", "r":3.8, "c":800}
];

func2(services, 15, 17, "c>=800"); // S3
func2(services, 11, 13, "r<=4"); // S3
func2(services, 10, 12, "name=S3"); // Sorry
func2(services, 15, 18, "r>=4.5"); // S1
func2(services, 16, 18, "r>=4"); // Sorry
func2(services, 13, 17, "name=S1"); // Sorry
func2(services, 8, 9, "c<=1500"); // S2
```

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Task 3:

There is a number sequence: 25, 23, 20, 21, 23, 21, 18, 19, 21, 19, 16, 17, ...

Find out the nth term in this sequence.

Note: Never change existing code.

Python

```
def func3(index):  
    # your code here  
func3(1) # print 23  
func3(5) # print 21  
func3(10) # print 16  
func3(30) # print 6
```

JavaScript

```
function func3(index){  
    // your code here  
}  
func3(1); // print 23  
func3(5); // print 21  
func3(10); // print 16  
func3(30); // print 6
```

Task 4:

Given available spaces for each car of a train, status bitmap, and number of incoming passengers, writing a procedure to find out the index of the most fitted car to serve passengers.

- Available Spaces: list/array containing number of available seats for each car.
- Status Bitmap: string containing only 0 or 1. 0 means the corresponding car can serve passengers for now.
- Passenger Number: number of incoming passengers.

We can assume all incoming passengers should be served in the same car.

Note: Never change existing code.

Python

```
def func4(sp, stat, n):  
    # your code here  
func4([3, 1, 5, 4, 3, 2], "101000", 2) # print 5  
func4([1, 0, 5, 1, 3], "10100", 4) # print 4  
func4([4, 6, 5, 8], "1000", 4) # print 2
```

JavaScript

```
function func4(sp, stat, n){  
    // your code here  
}  
func4([3, 1, 5, 4, 3, 2], "101000", 2); // print 5  
func4([1, 0, 5, 1, 3], "10100", 4); // print 4  
func4([4, 6, 5, 8], "1000", 4); // print 2
```